

BERA — Bio-Ecologic Resonance Architecture

Instrument ID: ERES-BERA-2026-001 **Version:** v1 — DRAFT **License:** CCAL v2.1 (CARE Commons Attribution License) **Author:** Joseph Allen Sprute, ERES Institute for New Age Cybernetics **ORCID:** 0000-0001-9946-3221 **Method:** H2C2H · MIEVM (Multi-Instrument Ensemble Validation Method) **Status:** Draft for MIEVM audit. Not deposited. Not load-bearing for any admit decision (see §8).

1. Purpose and scope

BERA specifies how a bio-ecologic system's **resonance** is measured from physiological observables and reduced to a structured **rating**, and how a separate **disruption index** $D(t)$ tracks coherence breakdown over time.

BERA is a *rating and tuning* architecture. It produces a quality measure, not an authorization. The decision of whether a consequence is permitted belongs to the gate (EAAP; see §3 and §8), never to BERA. This separation is the central design axiom and the rest of the document depends on it.

What BERA is **not**: it is not a clinical diagnostic, not a legally load-bearing instrument, and not — at this stage — an empirically established bridge to plasma physics. The cross-domain correspondence in §7 is a hypothesis under test, explicitly tagged, not a result.

2. The gate/rating split principle (load-bearing axiom)

Two functions must remain architecturally separate:

- **Gate** — a bounded decision: `(BIND / REFUSE / VEILED)`. Governs *whether* a consequence may occur. Deterministic, auditable, legally load-bearing. Implemented by EAAP (`(eaap_proof.py)`), not by BERA.
- **Rating** — a continuous, multi-dimensional measure (`(RATE)`, §6). Governs *how resonant* a state is. Tuning-tier, governance commentary.

The principle has two prohibitions, both of which are SOUND-tier correctness requirements:

1. A rating must **never** be silently promoted into a gate decision. A high score does not authorize anything.
2. A gate decision must **never** be derived by thresholding a scalar-collapsed rating. Authorization comes from the conjunctive gate factors, not from a number crossing a line.

The failure mode this prevents: a "score" quietly becoming a permission. BERA exists in part to make that failure structurally impossible by keeping the rating on its own side of the wall.

3. Relationship to the gate (EAAP)

BERA and EAAP are companion instruments. EAAP answers *admit/refuse* through four conjunctive Proof-of-Resonance factors over strong Kleene three-valued logic, reaching an effect only on `BIND`. BERA answers *how resonant* through the indices in §5 and the RATE vector in §6. BERA may inform tuning of EAAP thresholds as governance commentary; it may not bind a consequence. The receipt chain that records a gate decision (EAAP) and the rating that describes a state (BERA) are distinct records and must not be conflated.

4. Observables and measurement window

BERA derives its indices from non-invasive physiological and environmental channels:

- **HRV** — heart-rate variability (cardiac autonomic balance)
- **EDA** — electrodermal activity (sympathetic arousal)
- **Respiration** — rate and depth (respiratory coherence)
- **VOC** — volatile organic compounds (the ecologic/odor channel; cf. FAVORS "ODOR canonical")

Observation proceeds over a defined **window** (canonical test fixtures use a 30-second window; the operational window length is a tunable parameter, not a fixed constant). Each observable channel is sampled, normalized, and contributes to one or more indices in §5.

5. Resonance indices

BERA reduces the windowed observables to a small set of named indices. The canonical encoding fixes the field names and float precision (six decimal places, half-even rounding, per the EAAP/CRYPTO encoding rules):

```
{"ari": <0..1>, "eri": <0..1>, "rci": <0..1>, "rhc": <0..1>,  
  "sensor_spec": "<sensor id/version>", "window_seconds": <int>}
```

Index	Canonical name	Role
RHC	Resonant Harmony Cycle (LOCKED — never "Resonant Harmony Cybernetics")	Cyclic coherence across the window
ARI	[REQUIRES JAS CANONICAL CONFIRMATION] — proposed: Autonomic Resonance Index	Autonomic-channel resonance (HRV/EDA-weighted)
ERI	[REQUIRES JAS CANONICAL CONFIRMATION] — proposed: Ecologic Resonance Index	Bio-ecologic / environmental coupling (VOC-weighted)
RCI	[REQUIRES JAS CANONICAL CONFIRMATION] — proposed: Respiratory Coherence Index	Respiratory-channel coherence

Open item (canonical lock required): The expansions for ARI, ERI, and RCI above are *proposals only*. They are not locked. v2 must replace them with your confirmed canonical expansions before any external citation, consistent with ERES lock discipline.

These indices are the same four that feed value accrual under the Meritcoin / Proof-of-Resonance loop ("not mining — tuning"). Within BERA they are inputs to the RATE vector; their accrual role is downstream and out of scope for this instrument.

6. The RATE vector

The indices compose into **RATE**, a **seven-dimensional** vector ($R_1..R_7$). Per the canonical interlock:

$$(CBGMODD \times FAVORS) + BERA = RATE$$

Scalar collapse of RATE is PROHIBITED. RATE is reported across all seven dimensions; reducing it to a single number discards the dimensional structure the gate/rating split depends on (§2, prohibition 2). A dimension whose evidence is unresolvable in context is held in a **VEILED-R** state rather than coerced to a value (cf. CyberRAVE: Cybernetic Ratings Abolishing Veiled Exchanges). RATE computation uses integer cross-multiplication, not floating-point, so the result is deterministic and replayable across platforms.

7. The D(t) disruption index

D(t) is a **time-resolved disruption index**, normalized to $[0, 1]$, that tracks the rate of coherence loss across the observable channels within and across windows. Low D(t) = coherent/resonant; rising D(t) = approaching coherence breakdown; D(t) near 1 = disruption onset.

D(t) is the instrument under test in the **ERES-PELLIS Protocol**, where physiological coherence is stressed by a parametric acoustic stimulus across A-B-A-B-A-B blocks and D(t) is observed near onset.

HYPOTHESIZED (conceptual, not engineered): The working hypothesis is that the *functional form* of D(t) near coherence breakdown in a physiological system is shared with the functional form of spectral collapse near a disruption in a confined fusion plasma (the BERA-PSO correspondence). This is a hypothesis to be tested, not a demonstrated result, and it must be tagged as such wherever it appears. It was previously a SOUND-tier failure to describe this correspondence as "directly engineered"; it is a conceptual hypothesis only.

The architecture supports the test (it defines D(t) and the observation protocol); it does not assert the correspondence holds.

8. Position in the ERES instrument stack

BERA occupies the **tuning** tier of the three-instrument governance stack:

1. **Deterministic reference implementation (EAAP gate)** — legally load-bearing (Daubert-relevant).
2. **MIEVM-LLM ensemble** — governance commentary only.
3. **BERA** — governance tuning only.

Consequence-bearing authorization flows through tier 1. BERA (tier 3) tunes and describes; it does not authorize. Any document or claim that places BERA in a load-bearing decision role is out of specification.

9. Open items for v2 (MIEVM audit targets)

1. **Canonical lock** of ARI, ERI, RCI expansions (§5) — required before external citation.
2. **D(t) functional form** — the explicit normalized definition (rate-of-coherence-loss formula and onset behavior), currently described structurally only.
3. **Window length** — confirm the operational default vs. the 30 s test fixture.
4. **RATE dimension map** — the explicit $R_1..R_7$ assignment and the VEILED-R threshold rule.
5. **BERA-PSO** — keep HYPOTHESIZED tag; define the specific observable that would confirm or falsify shared functional form under the ERES-PELLIS protocol.
6. **Pellis approval gate** — any deposit of BERA material touching the correspondence is gated on Pellis sign-off, consistent with the BSG2 series discipline.

Three Governing Principles (operationally binding): Don't hurt yourself. Don't hurt others. Build for generations to come.